

Hyr

Engineering a Connected AI Recruiting Platform

Candidate intelligence • Role intelligence • Structured AI interviews • Ranking & decisions

Engineering case study • 10K+ users

10K+
users served
Platform scale

80%
lower role setup time
Verified outcome

70%
lower screening time
Verified outcome

4
conceptual system layers
Architecture model

ENGINEERING NARRATIVE

Structured candidate + role context → automated interviews → ranked decision signals.

Designed the structured-output LLM engine, asynchronous JD-generation pipeline, multi-stage interview agents, candidate profiling, and ranking flow while keeping role, candidate, interview, and scoring context connected.

CONNECTED RECRUITING FLOW

Candidate context stays connected from profile creation through ranking.

01
Candidate profile
resume-derived context

02
Matched roles
automatic surfacing

03
Direct application
same workflow

04
AI interview
multi-stage agents

05
Evaluation / ranking
structured signals

ENGINEERING SCOPE

Candidate Intelligence

Role Intelligence

AI Interviews

Evaluation

Workflow

Distribution

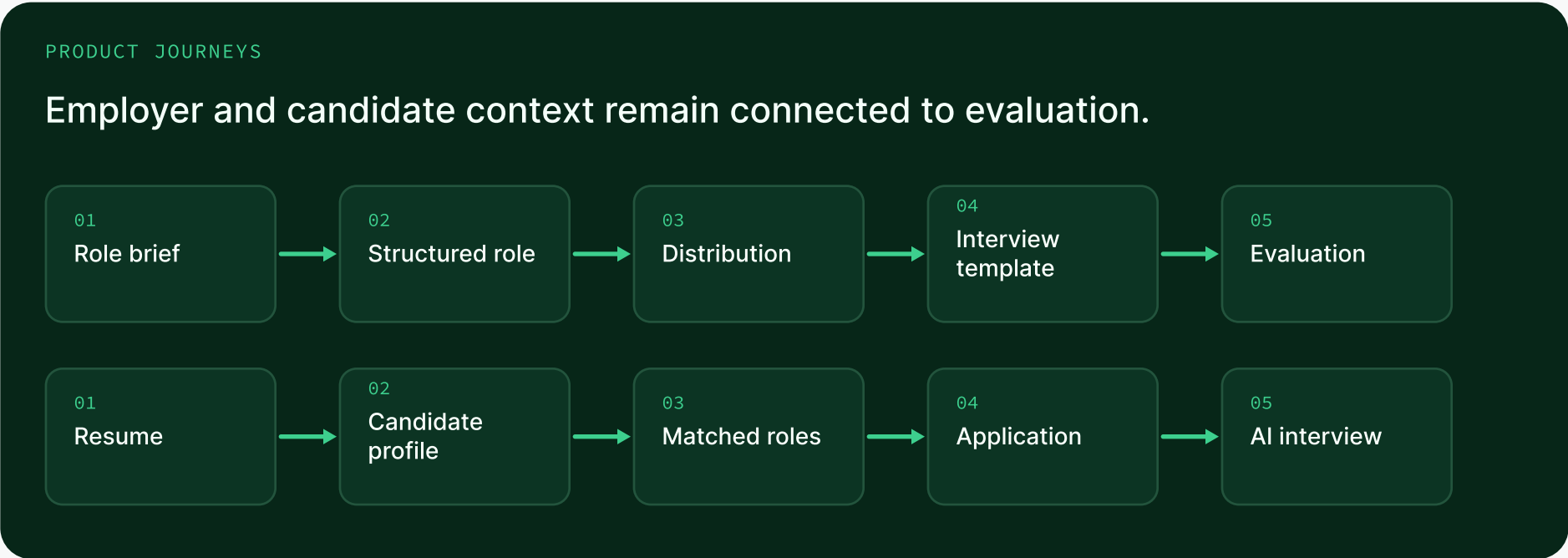
CORE TECHNOLOGIES

AI / LLM • structured-output LLM engine • OpenAI function calling

Candidate intelligence • spaCy • pgvector • skill embeddings • composite scoring • Backend • Celery • Pydantic

Product Flow & Engineering Ownership

Two connected journeys feed one structured evaluation and decision workflow.



OWNERSHIP LEDGER

BUILT	Structured-output LLM engine JD generation + evaluation	BUILT	Async Celery pipeline JD generation
IMPLEMENTED	Pydantic validation structured JD outputs	ENGINEERED	Multi-stage interview agents OpenAI function calling
BUILT	Candidate profiling spaCy + pgvector + scoring	CONNECTED	Candidate profile workflow match → apply → interview → evaluate
BUILT	Candidate ranking experience · skills · interview	SUPPORTED	Recruiting distribution career pages + external channels

CORE SYSTEM RESPONSIBILITIES

CONTEXT

Candidate intelligence

Profile outputs are reused across matching, interviewing, and evaluation.

CONTEXT

Role intelligence

Role outputs stay connected to downstream interview and scoring workflows.

AUTOMATION

Structured interviews

Multi-stage AI interview templates support technical, experience, coding, and Q&A rounds.

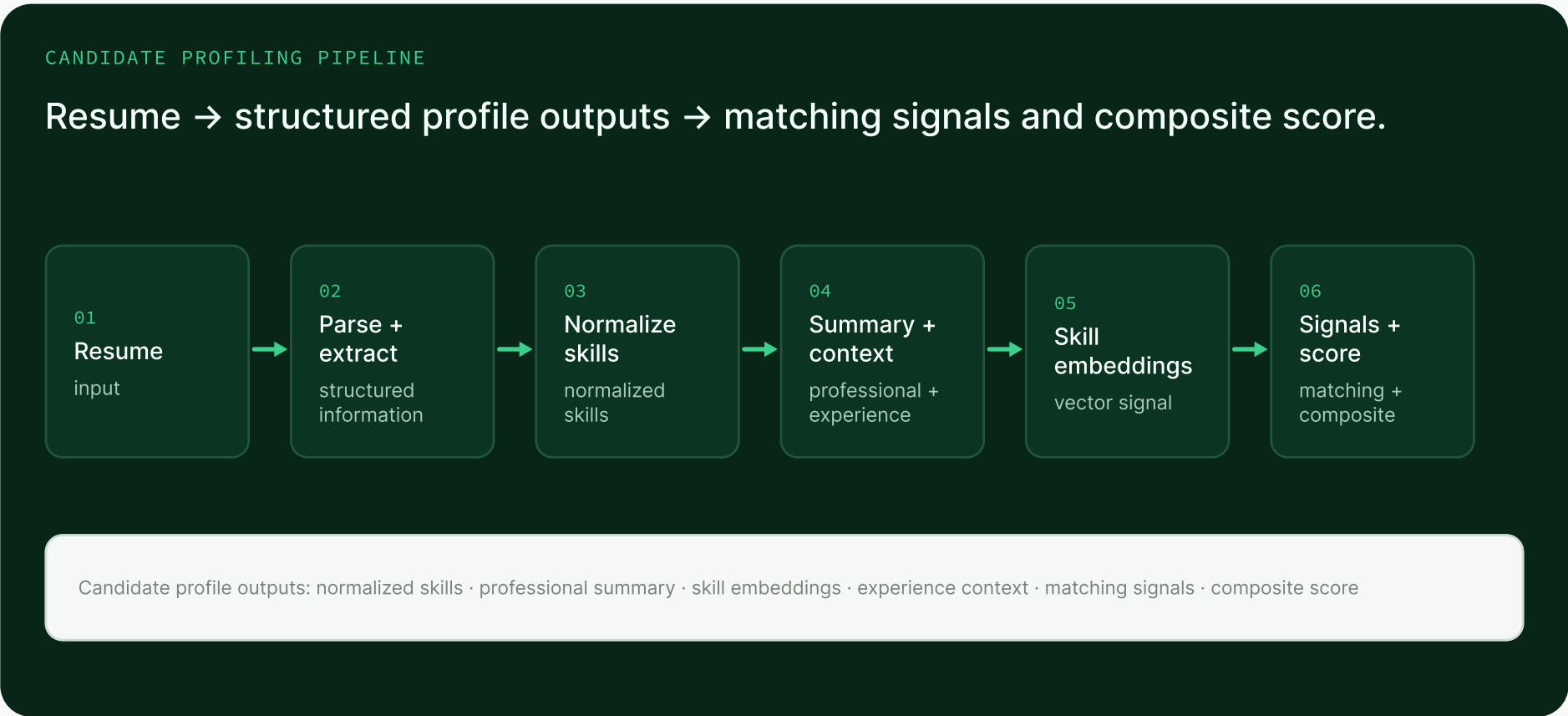
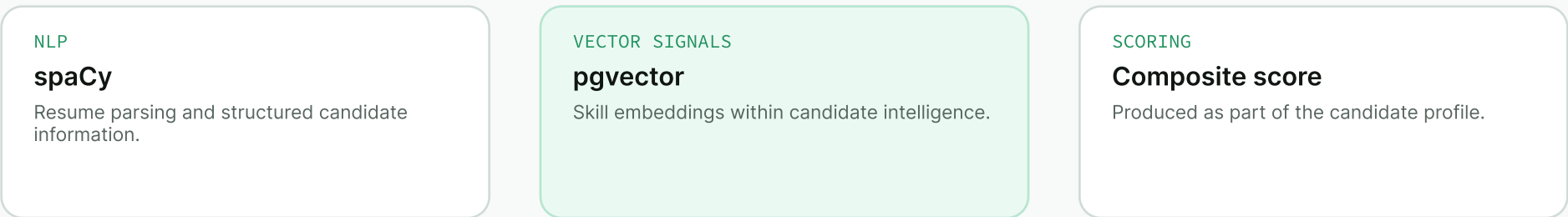
DECISIONS

Ranked signals

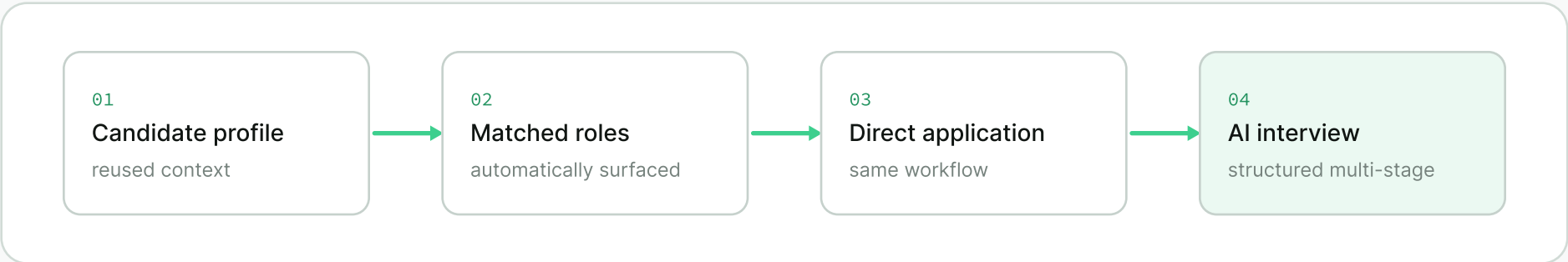
Ranking uses experience fit, skills match, and interview performance.

Candidate Intelligence & Matching

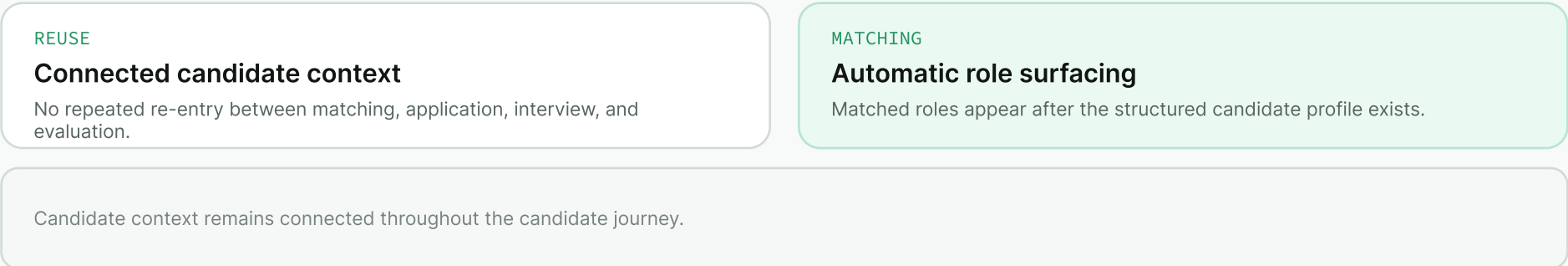
Resume-derived structured context becomes reusable matching and evaluation signal.



MATCHED-ROLE WORKFLOW



PROFILE BEHAVIOR



Role Intelligence & AI Interviews

Structured role generation feeds configurable multi-stage AI interviewing.

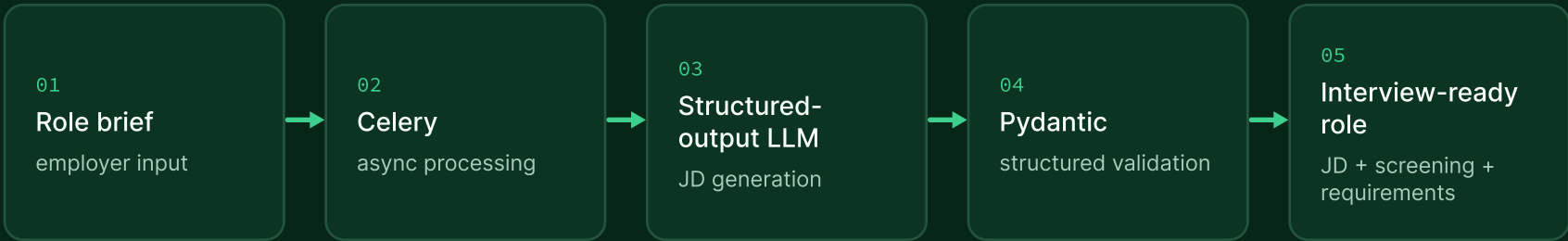
ENGINEERING FOCUS

Structured role generation and interview automation

Role creation produces AI-assisted descriptions, screening questions, and structured requirements that remain connected to downstream evaluation.

ROLE-GENERATION FLOW

Asynchronous structured-output generation with validation.



Verified outcome · role setup time reduced by 80%

SUPPORTED INTERVIEW STAGE TYPES



INTERVIEW SYSTEM

AGENTS

Multi-stage LLM interviews

Interview agents use OpenAI function calling.

CONTEXT

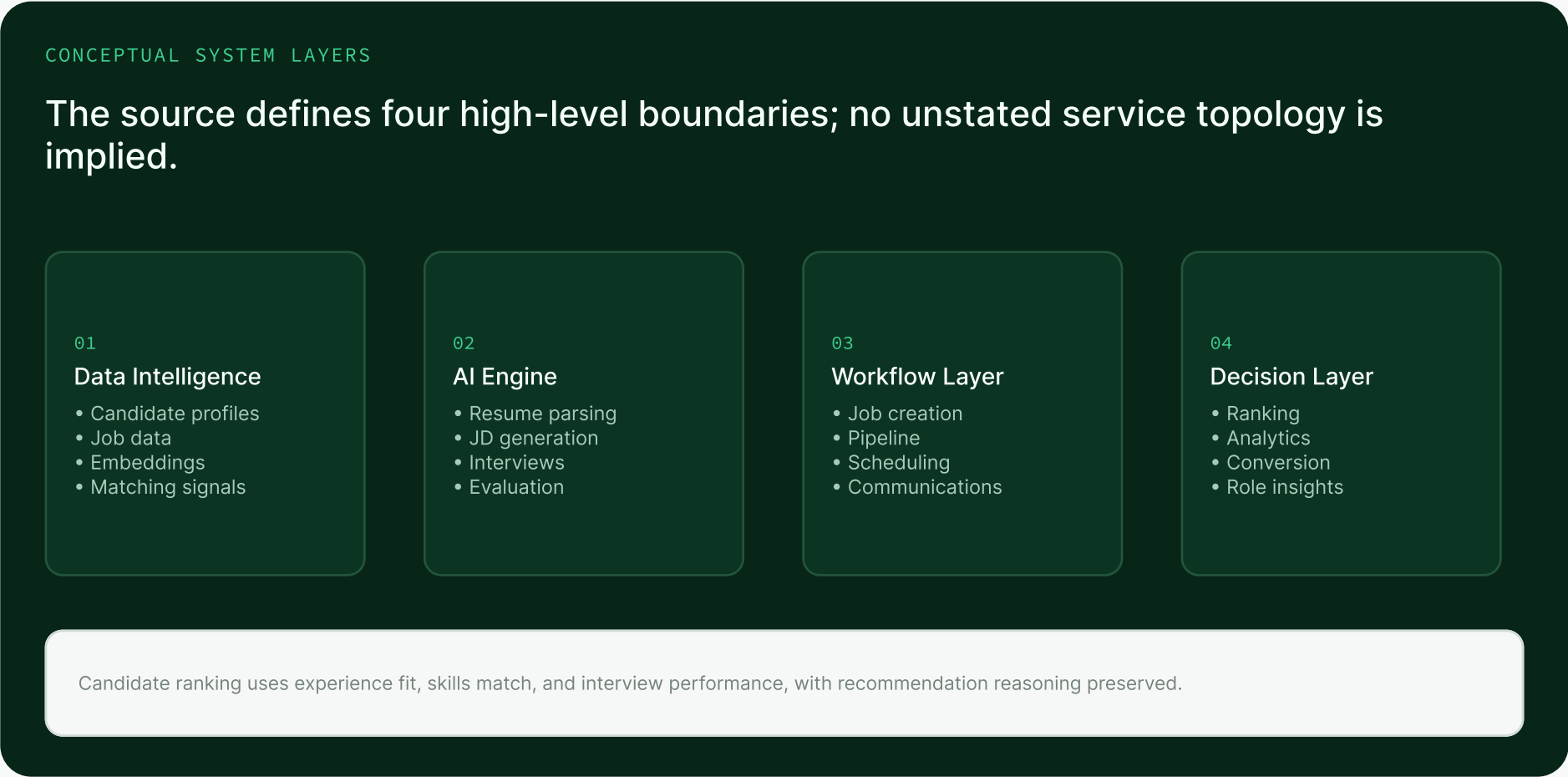
Continuity across stages

Role brief, interview flow, candidate context, and scoring logic stay connected.

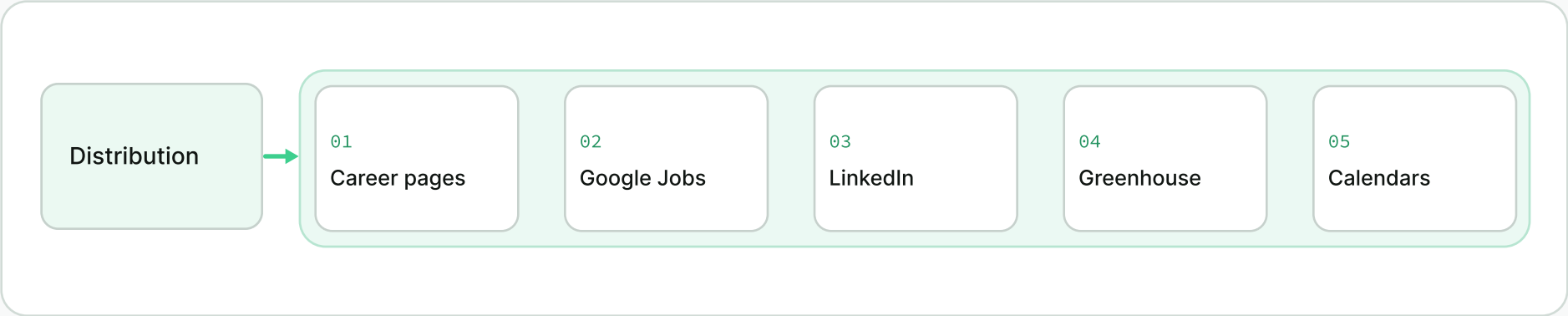
Verified outcome · structured AI interview automation reduced screening time by 70%.

Four-Layer System Architecture

Conceptual boundaries connect structured data, AI capabilities, workflow operations, and decisions.



CONNECTED RECRUITING CHANNELS



DECISION-LAYER OUTPUTS



Results, Engineering Decisions & Evidence

Verified quantitative outcomes plus the implementation choices explicitly established by the source.

VERIFIED RESULTS

SYSTEM / SCALE	RESULT	FRAMING
Platform scale	10K+ users	Verified scale
Role setup workflow	80% reduction in setup time	Verified outcome
Structured AI interviews	70% reduction in screening time	Verified outcome

ENGINEERING DECISIONS

STRUCTURE

Structured outputs + Pydantic
The role-generation workflow enforces structured LLM outputs with validation.

ASYNC

Celery JD generation
Job-description generation runs through an asynchronous Celery pipeline.

INTERVIEWS

Function-calling agents
Multi-stage LLM interview agents use OpenAI function calling.

RANKING

Three evaluation dimensions
Experience fit · skills match · interview performance.

PUBLICATION EVIDENCE BOUNDARY

Public artifact boundary: use only named technologies, established flows, verified metrics, and the four conceptual layers. No unstated model names, scoring weights, cloud/API/frontend stack, deployment topology, or proprietary prompt logic.

10K+ users

80% role setup

70% screening

High-level architecture